The hardest real VCAA questions

All 32 questions from the hardest third of the 2023–2025 Methods exams, ranked by how Victorian students actually scored on them, grouped by topic with the hardest first. Each one comes with the answer, the marking points and how the state went. A topic with nothing in the hardest third gets the hardest question there is on it, marked ★.

How to use it

- Work each question before reading the answer, then annotate the page with what caught you out.
- In Exam 2, go to the topic in the contents and use the marking points as a model of what earns each mark.
- The percentage next to each question is the share of Victorian students who got it right (multiple choice) or the average share of marks (written). Low numbers show where the marks are lost.
- Add your own notes pages, and tab the topics so you can find them fast.

VCAA rules for a bound reference (check the current year's list)

- It must be in book format, A4 or smaller when closed.
- No pages, or parts of pages, that can come out during the exam: loose pages are removed and reported as a possible breach.
- Anything you add must be firmly attached, e.g. with glue, tape or staples.
- Get it properly bound. Ask your school whether they accept spiral or ring binding before relying on it.

From the VCAA's authorised materials and equipment notes for VCE external assessments. Your school may have its own bound reference policy too.

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Not covered by any real 2023–2025 question in this set: power functions, limits and continuity, sampling. Revise those from the guide's topic pages.

TOPIC

Functions: domain, range, hybrids & modulus

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2023 VCAA Exam 2 · Section A · Question 3 ★

47% correct

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Domain of the sum of two functions with domains [−2, 3) and (−1, 5]

Open the question in the official exam PDF (p. 3) ↗

Answer: E · 47% correct · A 22% · B 12% · C 11% · D 7% · E 47%

The sum's domain is the intersection: $(-1, 3)$.

TOPIC

Polynomials: factor theorem, cubics & quartics

2 questions, hardest first.

2024 VCAA Exam 2 · Section A · Question 17

27% correct

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Output of an algorithm that searches for integer roots of a cubic

Open the question in the official exam PDF (p. 9) ↗

Answer: D · 27% correct · A 14% · B 26% · C 32% · D 27%

The roots are 3, −3 and 2. The loop counts c down from 18, checking c then −c, so it prints 3, −3, 2.

2023 VCAA Exam 2 · Section A · Question 19

32% correct

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Values of k giving one positive and one negative root of a quadratic

Open the question in the official exam PDF (p. 9) ↗

Answer: D · 32% correct · A 27% · B 11% · C 13% · D 32% · E 16%

Opposite signs need the product of the roots to be negative: $4k^2 - \frac{9}{4} < 0$, so $-\frac{3}{4} < k < \frac{3}{4}$.

TOPIC

Transformations of graphs

3 questions, hardest first.

2025 VCAA Exam 2 · Section A · Question 20

36% correct

© VCAA · full paper (p. 12) ↗

Which transformation sequence does not map a^x to a^{2x+2}

Open the question in the official exam PDF (p. 12) ↗

Answer: C · 36% correct · A 24% · B 18% · C 36% · D 21%

Apply each sequence in order. The sequence “dilate by a from the x-axis, dilate by ½ from the y-axis, then translate 1 right” gives a^{2x-1}.

2025 VCAA Exam 2 · Section A · Question 15

44% correct

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Image of the point (1, 3) on $y = 1 - g(2x - 3)$

Open the question in the official exam PDF (p. 9) ↗

Answer: B · 44% correct · A 25% · B 44% · C 22% · D 8%

You need $2x - 3 = 1$, so $x = 2$ and $y = 1 - 3 = -2$.

2024 VCAA Exam 2 · Section A · Question 13

45% correct

© VCAA · full paper (p. 7) ↗

Local minimum of $f(x) = \frac{x}{2} + \frac{2}{x}$ after dilation ×3 from the y-axis and a shift down 1

Open the question in the official exam PDF (p. 7) ↗

Answer: A · 45% correct · A 45% · B 24% · C 17% · D 14%

f has its minimum at (2, 2). The dilation gives (6, 2), and the shift gives (6, 1).

TOPIC

Exponential & logarithmic functions

2 questions, hardest first.

2024 VCAA Exam 1 · Question 6

state average 38% of marks

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Solve $2\log_3(x-4)+\log_3(x)=2$
Open the question in the official exam PDF (p. 7) ↗

Solve for x (4 marks · state average 1.5 / 4, 38%)

- Combines: $\log_3\left(x(x-4)^2\right)=2$
- $x^3-8x^2+16x-9=0$
- Factorises: $(x-1)(x^2-7x+9)=0$
- Only $x=\frac{7+\sqrt{13}}{2}$ satisfies $x>4$

Answer / notes: $x=\frac{7+\sqrt{13}}{2}$. Reject $x=1$ and $\frac{7-\sqrt{13}}{2}$ because the log needs x > 4.

2023 VCAA Exam 2 · Section B · Question 3

state average 44% of marks

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$g(x)=2^x+5$: limit, derivative, tangents through the origin; $h(x)=2^x-x^2$: inflection, decreasing interval, Newton's method
Open the question in the official exam PDF (p. 16) ↗

a. $\lim_{x\to-\infty} g(x)$ (1 mark · state average 0.7 / 1, 70%)

- 5

Answer / notes: 5
b. k with $g'(x)=k\times 2^x$ (1 mark · state average 0.9 / 1, 90%)

- $k=\log_e 2$

Answer / notes: $k=\log_e 2$
c.i. Tangent at $(a,g(a))$ (1 mark · state average 0.5 / 1, 50%)

- $y=2^a\log_e 2\,(x-a)+2^a+5$

Answer / notes: Point–gradient form.
c.ii. Tangent to g through the origin (3 d.p.) (2 marks · state average 0.6 / 2, 30%)

- Sets $0=-a\,2^a\log_e 2+2^a+5$, giving $a\approx 2.618$
- $y\approx 4.255x$

Answer / notes: Substitute (0, 0) into the general tangent, solve for a, then find the gradient.
d. Point of inflection of h (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $\approx(2.06,-0.07)$

Answer / notes: $h''(x)=(\log_e 2)^2 2^x-2=0$
e. Largest interval on which h is strictly decreasing (2 d.p.) (1 mark · state average 0.4 / 1, 40%)

- $[0.49,3.21]$

Answer / notes: Between the two roots of h′, with square brackets (the endpoints are included).
f. Newton's method from $x_0=0$: x_1,x_2,x_3 (3 d.p.) (2 marks · state average 1.2 / 2, 60%)

- $x_1\approx -1.443$
- $x_2\approx -0.897,\ x_3\approx -0.773$

Answer / notes: −1.443, −0.897, −0.773

g. Why a solution of $\log_e(2)2^x - 2x = 0$ is a bad starting point (1 mark · state average 0.2 / 1, 20%)

- That equation is $h'(x) = 0$, so the tangent is horizontal and Newton's formula divides by zero

Answer / notes: A stationary point makes the tangent horizontal, so it never meets the x-axis.
h. Positive n for which $n^x - x^n$ has a local minimum on the x-axis (2 marks · state average 0.2 / 2, 10%)

- Requires $f(x) = 0$ and $f'(x) = 0$ at the same x, giving $x = \frac{n}{\log_e n}$
- $n = e$

Answer / notes: $n = e$: the curve $e^x - x^e$ touches the axis at x = e. An exact answer was required.

TOPIC

Circular (trigonometric) functions

3 questions, hardest first.

2023 VCAA Exam 2 · Section A · Question 18

29% correct

© VCAA · full paper (p. 9) ↗

Number of local minima of $\sin(ax)$ on $[-a\pi, a\pi]$

Open the question in the official exam PDF (p. 9) ↗

Answer: E · 29% correct · A 12% · B 9% · C 22% · D 26% · E 29%

Interval length $2a\pi \div$ period $\frac{2\pi}{a} = a^2$ periods, each with one minimum.

2023 VCAA Exam 2 · Section A · Question 9

42% correct

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Value of a that makes a tan/sin hybrid continuous and smooth at x = 2π

Open the question in the official exam PDF (p. 5) ↗

Answer: C · 42% correct · A 7% · B 9% · C 42% · D 31% · E 10%

Continuity needs $\sin(2\pi a) = 0$. Smoothness needs $a\cos(2\pi a) = \frac{1}{2}$ (the gradient of $\tan\frac{x}{2}$ there). So $a = -\frac{1}{2}$.

2023 VCAA Exam 2 · Section B · Question 2

state average 45% of marks

© VCAA · full paper (p. 12) ↗

Observation wheel modelled by h(t) = −60cos(bt) + c; then it stops and restarts at double speed

Open the question in the official exam PDF (p. 12) ↗

a. Show $b = \frac{\pi}{15}$ and $c = 75$ (2 marks · state average 1.5 / 2, 75%)

- Period 30: $\frac{2\pi}{b} = 30 \Rightarrow b = \frac{\pi}{15}$
- $h(0) = -60 + c = 15 \Rightarrow c = 75$

Answer / notes: Use the period and the lowest height.
b. Average height from A (bottom) to B (level with the centre) (2 d.p.) (2 marks · state average 0.9 / 2, 45%)

- $\frac{1}{7.5}\int_0^{7.5} h(t)\,dt$
- $= 75 - \frac{120}{\pi} \approx 36.80$ m

Answer / notes: A quarter turn takes 7.5 minutes. ≈ 36.80 m
c. Average rate of change of height from A to B (1 mark · state average 0.5 / 1, 50%)

- $\frac{75-15}{7.5} = 8$ m/min

Answer / notes: 8 m/min
d.i. k and m in the piecewise model (1 mark · state average 0.4 / 1, 40%)

- $k = 135,\ m = 2$

Answer / notes: It stops at the top (135 m) and restarts at double speed.
d.ii. All possible values of n (2 marks · state average 0.4 / 2, 20%)

- Continuity at t = 20: $h(40+n) = 135$, so $\cos\left(\frac{\pi(40+n)}{15}\right) = -1$
- $n = 30j - 25,\ j \in Z$

Answer / notes: $n = 5 + 30j$ for integer j. A general solution was required.

d.iii. Sketch r with the coordinates of the endpoints (3 marks · state average 1.2 / 3, 40%)

- Rising cosine from $(0, 15)$ to $(15, 135)$
- Flat from $(15, 135)$ to $(20, 135)$
- Half a (faster) cycle down to $(27.5, 15)$

Answer / notes: Three pieces joined continuously, with endpoints (0, 15) and (27.5, 15).

TOPIC

Composite, inverse & combined functions

4 questions, hardest first.

2023 VCAA Exam 2 · Section A · Question 20

30% correct

© VCAA · full paper (p. 9) ↗

Largest interval where both $f \circ g$ and $g \circ f$ exist (log and sine)

Open the question in the official exam PDF (p. 9) ↗

Answer: A · 30% correct · A 30% · B 19% · C 26% · D 14% · E 9%

$g \circ f$ needs $x > -\frac{1}{\sqrt{2}}$. $f \circ g$ needs $\sin x > -\frac{1}{\sqrt{2}}$, i.e. $x < \frac{5\pi}{4}$ here. Combined: $\left(-\frac{1}{\sqrt{2}}, \frac{5\pi}{4}\right)$.

2023 VCAA Exam 2 · Section B · Question 5

state average 35% of marks

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$g(x) = \frac{1}{2}f(2-x)$ with $f(x) = e^x + e^{-x}$: transformations, restricted inverses, areas with inverses, a family of curves

Open the question in the official exam PDF (p. 21) ↗

a. Complete the sequence of transformations from f to g (2 marks · state average 1.2 / 2, 60%)

- Reflection in the y-axis
- Then translation 2 units right (or just translate 2 right, since f is even)

Answer / notes: Dilation ½ from the x-axis, reflect in the y-axis, then translate right 2. Order matters.
b. Domain and range of the inverse of the increasing branch g_1 (2 marks · state average 1.0 / 2, 50%)

- Domain $[1, \infty)$
- Range $[2, \infty)$

Answer / notes: The domain and range of the inverse are swapped from g_1 (turning point (2, 1)).
c.i. Coordinates of P and Q (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $P \approx (1.27, 1.27),\ Q \approx (4.09, 4.09)$

Answer / notes: Solve g(x) = x. Both points lie on y = x.
c.ii. Area bounded by g and the two inverses (2 d.p.) (2 marks · state average 0.6 / 2, 30%)

- $2\int_{1.27}^{4.09} (x - g(x))\,dx$
- ≈ 5.56

Answer / notes: Use the symmetry about y = x: double the area between y = x and g. Forgetting the 2 gives 2.78.
d. Integer n such that the turning point of h always lies on $y = 2x^n$ (1 mark · state average 0.1 / 1, 10%)

- $n = -1$

Answer / notes: The turning point is $\left(k, \frac{2}{k}\right)$, and $\frac{2}{k} = 2k^{-1}$.
e. Smallest k for which h meets the inverse of h_1 (2 d.p.) (1 mark · state average 0.0 / 1, 0%)

- $k \approx 1.27$

Answer / notes: Find the k where the curves just touch (a tangency condition). The state averaged 0.0/1.
f. Area between h and the inverse of h_1 when k = 5 (2 d.p.) (2 marks · state average 0.3 / 2, 15%)

- Intersections at x ≈ 1.45 and x ≈ 8.78
- Area ≈ 43.91

Answer / notes: Integrate $|h - h_1^{-1}|$ between the two intersection points.

2025 VCAA Exam 2 · Section A · Question 13

45% correct

© VCAA · full paper (p. 8) ↗

Graph of $g(f(x))$ given sketches of a line f and an even function g

Open the question in the official exam PDF (p. 8) ↗

Answer: C · 45% correct · A 12% · B 17% · C 45% · D 25%

f is $y = -x$ and g is even, so $g(-x) = g(x)$. The composite looks just like g.

2024 VCAA Exam 2 · Section B · Question 5

state average 45% of marks

© VCAA · full paper (p. 24) ↗

Composites of $\sin x$ and $\sin 2x$: maxima, ranges, derivatives, stationary points, area between them, domain of a composite

Open the question in the official exam PDF (p. 24) ↗

a.i. Local maximum of $g \circ f$ in $[0, \frac{\pi}{2}]$ (1 d.p.) (1 mark · state average 0.7 / 1, 70%)

- $(0.9, 1)$

Answer / notes: $\sin(2\sin x) = 1$ when $\sin x = \frac{\pi}{4}$: $x \approx 0.9$.
a.ii. Range of $g \circ f$ on $[0, 2\pi]$ (1 mark · state average 0.6 / 1, 60%)

- $[-1, 1]$

Answer / notes: $2\sin x$ covers [−2, 2], which contains $\pm\frac{\pi}{2}$, so the full $[-1, 1]$.
b.i. Derivative of $f \circ g$ (1 mark · state average 0.8 / 1, 80%)

- $2\cos(2x)\cos(\sin 2x)$

Answer / notes: Chain rule.
b.ii. Show $\cos(\sin 2x) = 0$ has no real solutions (2 marks · state average 0.8 / 2, 40%)

- It would need $\sin 2x = \frac{\pi}{2} + k\pi$
- But $|\sin 2x| \leq 1 < \frac{\pi}{2}$, so there are no solutions

Answer / notes: The sine can't reach ±π/2.
b.iii. x-values of the stationary points of $f \circ g$ on $[0, 2\pi]$ (1 mark · state average 0.5 / 1, 50%)

- $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

Answer / notes: Only $\cos 2x = 0$ contributes (by part ii).
b.iv. Range of $f \circ g$ (1 mark · state average 0.4 / 1, 40%)

- $[-\sin 1, \sin 1]$

Answer / notes: $\sin(\sin 2x)$ with $\sin 2x \in [-1, 1]$
c.i. Single definite integral for the area between the two composites on [0, 2π] (1 mark · state average 0.3 / 1, 30%)

- $\int_0^{2\pi} |\sin(\sin 2x) - \sin(2\sin x)|\, dx$

Answer / notes: Use the absolute value so one integral covers every region.
c.ii. That area (2 d.p.) (1 mark · state average 0.5 / 1, 50%)

- ≈ 4.97

Answer / notes: ≈ 4.97
d. x in $(0, 2\pi)$ for which $f_1 \circ g$ is defined, f_1 having domain $(0, 2\pi)$ (2 marks · state average 0.3 / 2, 15%)

- Requires $\sin 2x > 0$ (range of g inside the domain of f_1)
- $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

Answer / notes: The range of g must sit inside (0, 2π), so $0 < \sin 2x$.

TOPIC

Functional equations, odd & even functions

1 question, hardest first.

2025 VCAA Exam 1 · Question 9

state average 27% of marks

© VCAA · full paper (p. 12) ↗

Solving $\frac{w^2}{(x-1)^2} = (x-w)^2$; the minimum of $(x-1)(x-w)$; when there are exactly three solutions

Open the question in the official exam PDF (p. 12) ↗

a. Four solutions when w = −3 (3 marks · state average 1.0 / 3, 33%)

- Reduces to $(x+3)(x-1) = \pm 3$
- $+3$: $x^2 + 2x - 6 = 0 \Rightarrow x = -1 \pm \sqrt{7}$
- -3: $x^2 + 2x = 0 \Rightarrow x = 0, -2$

Answer / notes: $x = -2, 0, -1 \pm \sqrt{7}$

b.i. Minimum point of $y = (x-1)(x-w)$, $w > 0$ (2 marks · state average 0.8 / 2, 40%)

- $x = \frac{w+1}{2}$
- $y = -\frac{(w-1)^2}{4}$

Answer / notes: $\left(\frac{w+1}{2}, -\frac{(w-1)^2}{4}\right)$

b.ii. Positive w giving exactly three solutions (2 marks · state average 0.1 / 2, 5%)

- $(x-1)(x-w) = w$ always gives 2 solutions, so $(x-1)(x-w) = -w$ must touch the minimum: $-w = -\frac{(w-1)^2}{4}$
- $w^2 - 6w + 1 = 0 \Rightarrow w = 3 \pm 2\sqrt{2}$

Answer / notes: $w = 3 \pm 2\sqrt{2}$. The state averaged 0.1/2.

TOPIC

Simultaneous equations & parameters

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2023 VCAA Exam 2 · Section A · Question 4 ★

55% correct

© VCAA · full paper (p. 3) ↗

Values of k giving infinitely many solutions

Open the question in the official exam PDF (p. 3) ↗

Answer: B · 55% correct · A 15% · B 55% · C 10% · D 14% · E 4%

The determinant $k^2 + k - 20 = 0$ gives k = −5 or 4. Only k = −5 makes the equations identical. k = 4 gives parallel lines.

TOPIC

Newton's method, bisection & pseudocode

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2023 VCAA Exam 2 · Section A · Question 13 ★

52% correct

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Result of three Newton's method iterations on $x^3 + 3x - 3$ from $x_0 = 1$

Open the question in the official exam PDF (p. 6) ↗

Answer: C · 52% correct · A 16% · B 22% · C 52% · D 7% · E 3%

1 → 0.83333 → 0.81785 → 0.81773. Count exactly three iterations.

TOPIC

Differentiation rules

3 questions, hardest first.

2025 VCAA Exam 2 · Section A · Question 16

18% correct

© VCAA · full paper (p. 9) ↗

Conditions on a, b if $h(x) = a\log_e(bx)$ has $h'(x) > 0$ everywhere

Open the question in the official exam PDF (p. 9) ↗

Answer: D · 18% correct · A 20% · B 16% · C 46% · D 18%

$h'(x) = \frac{a}{x}$, with domain $bx > 0$. If b > 0 then x > 0 and you need a > 0. If b < 0 then x < 0 and you need a < 0. Either way ab > 0.

2023 VCAA Exam 2 · Section A · Question 11

22% correct

© VCAA · full paper (p. 5) ↗

Gradient of f × g at x = −2 from given values

Open the question in the official exam PDF (p. 5) ↗

Answer: E · 22% correct · A 6% · B 13% · C 8% · D 51% · E 22%

Product rule: $3 \times 8 + (-7) \times 2 = 10$.

2024 VCAA Exam 2 · Section A · Question 16

36% correct

© VCAA · full paper (p. 8) ↗

Gradient of $y = \sqrt{f(x)}$ at x = 4 given f(4) = 25 and $f'(4) = 15$

Open the question in the official exam PDF (p. 8) ↗

Answer: D · 36% correct · A 22% · B 14% · C 27% · D 36%

$\frac{f'}{2\sqrt{f}} = \frac{15}{10} = \frac{3}{2}$.

TOPIC

Applications of differentiation

4 questions, hardest first.

2023 VCAA Exam 2 · Section A · Question 14

29% correct

© VCAA · full paper (p. 7) ↗

Number of tangents to a cubic that pass through its positive x-intercept
Open the question in the official exam PDF (p. 7) ↗

Answer: D · 29% correct · A 6% · B 33% · C 22% · D 29% · E 10%

Set up the tangent at a general point and require it to pass through $\left(\frac{1}{3}, 0\right)$. The resulting cubic in the tangent point has 3 real solutions.

2024 VCAA Exam 1 · Question 8

state average 34% of marks

© VCAA · full paper (p. 10) ↗

Cube-root function $g(x) = \sqrt[3]{x-k} + m$: y-intercept, gradient there, through the origin, a second point with equal gradient
Open the question in the official exam PDF (p. 10) ↗

a. Coordinates of the y-intercept P (1 mark · state average 0.6 / 1, 60%)

- $\left(0,\ m - \sqrt[3]{k}\right)$

Answer / notes: $P = \left(0, m - \sqrt[3]{k}\right)$ (the cube root of −k is $-\sqrt[3]{k}$).
b. Gradient of g at P in terms of k (2 marks · state average 0.8 / 2, 40%)

- $g'(x) = \frac{1}{3}(x-k)^{-2/3}$
- $g'(0) = \frac{1}{3k^{2/3}}$

Answer / notes: $\frac{1}{3\sqrt[3]{k^2}}$. Keep it positive: $(-k)^{2/3} = k^{2/3}$.
c. k in terms of m if the graph passes through the origin (1 mark · state average 0.5 / 1, 50%)

- $k = m^3$

Answer / notes: $m - \sqrt[3]{k} = 0 \Rightarrow k = m^3$
d. Coordinates of Q (≠ P) with the same gradient, graph through the origin (3 marks · state average 0.5 / 3, 17%)

- $(x-k)^{2/3} = k^{2/3} \Rightarrow x - k = \pm k$
- $x = 2k$ (x = 0 is P)
- $Q = (2m^3, 2m)$

Answer / notes: By the symmetry of the cube root about its inflection point (k, m): $Q = (2m^3, 2m)$. The state averaged 0.5/3.

2024 VCAA Exam 2 · Section A · Question 10

38% correct

© VCAA · full paper (p. 4) ↗

What must be true if $f'(2) < 0$ and $f'(4) > 0$
Open the question in the official exam PDF (p. 4) ↗

Answer: B · 38% correct · A 26% · B 38% · C 16% · D 20%

f' is continuous and changes sign, so f is not one-to-one on [0, 5] and has no inverse. The others might not hold.

2025 VCAA Exam 2 · Section B · Question 4

state average 45% of marks

© VCAA · full paper (p. 23) ↗

$f(x) = \sin x + 1$ on $[0, \frac{5\pi}{2}]$: values, periodicity, tangents, Newton's method, a family of tangents, matching a polynomial

Open the question in the official exam PDF (p. 23) ↗

a. $f\left(\frac{2\pi}{3}\right)$ (1 mark · state average 0.9 / 1, 90%)

- $\frac{\sqrt{3}}{2} + 1$

Answer / notes: $\frac{\sqrt{3}}{2} + 1$
b. Exact x with $f(x) = \frac{3}{2}$ (1 mark · state average 0.8 / 1, 80%)

- $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$

Answer / notes: $\sin x = \frac{1}{2}$. Three solutions lie in the domain.
c. k and the largest a with f(x + k) = f(x) on [0, a] (2 marks · state average 0.6 / 2, 30%)

- $k = 2\pi$
- Largest $a = \frac{\pi}{2}$

Answer / notes: The period is 2π, and $x + 2\pi$ must stay within $\frac{5\pi}{2}$.
d. Tangent at $x = \frac{2\pi}{3}$ (1 mark · state average 0.7 / 1, 70%)

- $y = -\frac{x}{2} + \frac{\pi}{3} + \frac{\sqrt{3}}{2} + 1$

Answer / notes: Gradient $\cos\frac{2\pi}{3} = -\frac{1}{2}$.
e.i. Newton's method from $x_0 = \frac{2\pi}{3}$: x_2 to 1 d.p. (1 mark · state average 0.6 / 1, 60%)

- $x_2 \approx 5.2$

Answer / notes: $x_1 = \frac{2\pi}{3} + \sqrt{3} + 2 \approx 5.83$, then $x_2 \approx 5.2$.
e.ii. Draw the tangent at $x = x_1$ on the graph (1 mark · state average 0.3 / 1, 30%)

- Tangent at $x_1 \approx 5.83$ drawn, meeting the x-axis at $x_2 \approx 5.2$

Answer / notes: The tangent at x_1 should cut the x-axis at about 5.2.
f.i. Show the tangent at $x = p$ is $t(x) = \cos(p)(x - p) + \sin(p) + 1$ (2 marks · state average 1.1 / 2, 55%)

- Gradient $f'(p) = \cos p$ through $(p, \sin p + 1)$
- Correct point–gradient form rearranged

Answer / notes: $y - (\sin p + 1) = \cos p\,(x - p)$
f.ii. Minimum and maximum y-intercepts of t for $p \in (0, \frac{5\pi}{2})$ (2 marks · state average 0.5 / 2, 25%)

- y-intercept $= \sin p - p\cos p + 1$, derivative $p\sin p$, stationary at $\pi, 2\pi$
- Minimum $1 - 2\pi$, maximum $1 + \pi$

Answer / notes: At $p = 2\pi$: $1 - 2\pi$. At $p = \pi$: $1 + \pi$.
f.iii. p for which t has a unique x-intercept equal to f's x-intercept (2 d.p.) (2 marks · state average 0.4 / 2, 20%)

- Requires $t\left(\frac{3\pi}{2}\right) = 0$ with $p \neq \frac{3\pi}{2}$ (horizontal tangent excluded)
- $p \approx 2.38$ and $p \approx 7.04$

Answer / notes: Solve $\cos p\left(\frac{3\pi}{2} - p\right) + \sin p + 1 = 0$: p ≈ 2.38, 7.04.
g.i. Show c = 1 and d = 1 given $g(0) = f(0), g'(0) = f'(0)$ (2 marks · state average 1.4 / 2, 70%)

- $g(0) = d = f(0) = 1$
- $g'(0) = c = \cos 0 = 1$

Answer / notes: Direct substitution.
g.ii. Area between f and g on $[0, 2\pi]$ when they also match at 2π (2 d.p.) (2 marks · state average 0.7 / 2, 35%)

- Solves for $a = \frac{1}{2\pi^2}, b = -\frac{3}{2\pi}$
- Area ≈ 1.53

Answer / notes: Use $\int_0^{2\pi} |f - g|\,dx \approx 1.53$.

gmin b and r with $g(r) = f(r)$, $g'(r) = f'(r)$ when $a = 0$, $c = 1$ (2 marks · state average 0.57 / 2, 28%)

- Sets up $br^2 + r = \sin r$ and $2br + 1 = \cos r$
- $r = \pi,\ b = -\frac{1}{\pi}$

Answer / notes: $r = \pi$ gives $b = -\frac{1}{\pi}$ (the other solution lies outside the domain).

TOPIC

Optimisation (max/min problems)

2 questions, hardest first.

2025 VCAA Exam 2 · Section A · Question 19

14% correct

© VCAA · full paper (p. 12) ↗

Value of c when the shortest distance from $y = x + c$ to $y = \log_e(x - 1)$ is $\sqrt{2}$

Open the question in the official exam PDF (p. 12) ↗

Answer: D · 14% correct · A 35% · B 25% · C 24% · D 14%

The closest point on the curve has gradient 1: (2, 0). Distance $\frac{|2+c|}{\sqrt{2}} = \sqrt{2}$ gives c = 0 or −4, but c = −4 cuts the curve (distance 0), so c = 0.

2023 VCAA Exam 2 · Section A · Question 17

28% correct

© VCAA · full paper (p. 8) ↗

Volume of a cylinder cut from a sheet, in terms of x and y

Open the question in the official exam PDF (p. 8) ↗

Answer: B · 28% correct · A 6% · B 28% · C 26% · D 26% · E 13%

The circumference $y = 2\pi r$ and $h = x - 4r$, so $V = \pi r^2 h = \frac{\pi x y^2 - 2y^3}{4\pi^2}$.

TOPIC

Antidifferentiation

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2025 VCAA Exam 1 · Question 2 ★

state average 60% of marks

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Find $g(x)$ given $g'(x) = \frac{1}{2x+3}$ and $g(1) = 0$

Open the question in the official exam PDF (p. 3) ↗

Antidifferentiate and use the condition (2 marks · state average 1.2 / 2, 60%)

- $g(x) = \frac{1}{2}\log_e(2x+3) + c$
- $c = -\frac{1}{2}\log_e 5$, so $g(x) = \frac{1}{2}\log_e\left(\frac{2x+3}{5}\right)$

Answer / notes: $g(x) = \frac{1}{2}\log_e(2x+3) - \frac{1}{2}\log_e 5$. Don't forget the $\frac{1}{2}$ from the chain rule.

TOPIC

Definite integrals, areas & average value

2 questions, hardest first.

2025 VCAA Exam 2 · Section A · Question 17

38% correct

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Which graph fits $\int_1^2 f\,dx > \int_1^3 f\,dx$

Open the question in the official exam PDF (p. 10) ↗

Answer: A · 38% correct · A 38% · B 20% · C 17% · D 24%

The condition means $\int_2^3 f(x)\,dx < 0$, so f must be mostly below the axis on [2, 3].

2024 VCAA Exam 2 · Section A · Question 20

41% correct

© VCAA · full paper (p. 10) ↗

$\int_2^6 f(x)\,dx$ for a period-2 function with average value k on [0, 2]

Open the question in the official exam PDF (p. 10) ↗

Answer: C · 41% correct · A 20% · B 23% · C 41% · D 14%

Each period contributes 2k, and [2, 6] is two periods, so 4k.

TOPIC

Numerical integration: trapezium & rectangle rules

1 question, hardest first.

2024 VCAA Exam 1 · Question 7

state average 42% of marks

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$f(x) = x\sin(x)$: trapezium approximation, the range of f' on an interval, a stationary point, sketching f'
Open the question in the official exam PDF (p. 8) ↗

a. Trapezium rule, step $\frac{\pi}{3}$, on $[0, \pi]$ (3 marks · state average 1.5 / 3, 50%)

- Values: $f(0) = 0,\ f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}\pi}{6},\ f\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}\pi}{3},\ f(\pi) = 0$
- Correct trapezium formula applied
- $\frac{\sqrt{3}\pi^2}{6}$

Answer / notes: $\frac{\pi}{6}\left[0 + 2\left(\frac{\sqrt{3}\pi}{6} + \frac{\sqrt{3}\pi}{3}\right) + 0\right] = \frac{\sqrt{3}\pi^2}{6}$
b.i. Find $f'(x)$ (1 mark · state average 0.9 / 1, 90%)

- $\sin(x) + x\cos(x)$

Answer / notes: $f'(x) = \sin(x) + x\cos(x)$
b.ii. Range of f' over $\left[\frac{\pi}{2}, \frac{2\pi}{3}\right]$ (1 mark · state average 0.2 / 1, 20%)

- $\left[\frac{\sqrt{3}}{2} - \frac{\pi}{3},\ 1\right]$

Answer / notes: $f'\left(\frac{\pi}{2}\right) = 1$ and $f'\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{3}$ (decreasing between them).
b.iii. Hence verify a stationary point in that interval (1 mark · state average 0.1 / 1, 10%)

- f' is continuous and changes sign (1 > 0 and $\frac{\sqrt{3}}{2} - \frac{\pi}{3} < 0$), so $f'(x) = 0$ somewhere in the interval

Answer / notes: The range contains 0 and f′ is continuous, so it must hit zero. Only 12% got this.
c. Sketch $y = f'(x)$ on $[-\pi, \pi]$ with endpoints (3 marks · state average 1.1 / 3, 37%)

- Endpoints $(-\pi, \pi)$ and $(\pi, -\pi)$
- Odd-function shape through the origin with the given min (−1.1, −1.4) and max (1.1, 1.4)
- x-intercepts where f has turning points (≈ ±2.03)

Answer / notes: An odd curve from $(-\pi, \pi)$ down through ≈ −2 to the minimum, up through O to the maximum, then down through ≈ 2 to $(\pi, -\pi)$.

TOPIC

Probability: conditional & independent events

1 question, hardest first.

2024 VCAA Exam 2 · Section A · Question 11

40% correct

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Choosing 3 of 12 students: probability exactly two are from the front row of seven

Open the question in the official exam PDF (p. 5) ↗

Answer: B · 40% correct · A 12% · B 40% · C 10% · D 37%

$$\frac{\binom{7}{2}\binom{5}{1}}{\binom{12}{3}} = \frac{105}{220} = \frac{21}{44}$$

TOPIC

Discrete random variables

1 question, hardest first.

2023 VCAA Exam 2 · Section A · Question 12

29% correct

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Maximum possible mean of a pmf with parameter k

Open the question in the official exam PDF (p. 6) ↗

Answer: E · 29% correct · A 11% · B 14% · C 26% · D 18% · E 29%

All probabilities must lie in [0, 1], so $0 \leq k \leq \sqrt{5} - 2$. The mean $-3k^2 - 7k + 2$ is largest at k = 0, giving 2.

TOPIC

The binomial distribution

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2023 VCAA Exam 2 · Section A · Question 8 ★

49% correct

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Probability of at least one green in 8 draws with replacement

Open the question in the official exam PDF (p. 4) ↗

Answer: C · 49% correct · A 12% · B 14% · C 49% · D 16% · E 9%

$1 - \Pr(\text{no green}) = 1 - \left(\frac{m}{n+m}\right)^8$.

TOPIC

Continuous random variables (pdfs)

2 questions, hardest first.

2023 VCAA Exam 1 · Question 8

state average 32% of marks

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Queueing-time pdf $f(t) = kt(16 - t^2)$ on [0, 4]: find k, E(T), and a conditional probability

Open the question in the official exam PDF (p. 10) ↗

a. Show that $k = \frac{1}{64}$ (1 mark · state average 0.4 / 1, 40%)

- $k \int_0^4 (16t - t^3)\,dt = 64k = 1$

Answer / notes: $k\left[8t^2 - \frac{t^4}{4}\right]_0^4 = 64k = 1$

b. E(T) (2 marks · state average 0.8 / 2, 40%)

- $\frac{1}{64}\int_0^4 (16t^2 - t^4)\,dt$
- $= \frac{32}{15}$

Answer / notes: $\frac{1}{64}\left(\frac{1024}{3} - \frac{1024}{5}\right) = \frac{32}{15}$

c. $\Pr(T > 2 \mid T > 1)$ (3 marks · state average 0.7 / 3, 23%)

- $\Pr(T > 2) = \frac{9}{16}$
- $\Pr(T > 1) = \frac{225}{256}$
- Ratio $= \frac{16}{25}$

Answer / notes: $\frac{9/16}{225/256} = \frac{16}{25}$

2025 VCAA Exam 1 · Question 8

state average 44% of marks

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Continuous pdf: find k with Pr(X > k) = 9/16; integral of a transformed pdf

Open the question in the official exam PDF (p. 10) ↗

a. Find k such that $\Pr(X > k) = \frac{9}{16}$ (3 marks · state average 1.4 / 3, 47%)

- Sets up $\int_k^{4/3} \frac{3}{8}(4 - 3x)\,dx = \frac{9}{16}$ (or the complement)
- Gets $9k^2 - 24k + 7 = 0$
- $k = \frac{1}{3}$ (rejects $\frac{7}{3}$, outside the domain)

Answer / notes: $k = \frac{1}{3}$. The other root $\frac{7}{3}$ is outside $\left[0, \frac{4}{3}\right]$.

b. $\int_0^{4/3} h(x)\,dx$ for $h(x) = mf(x) + n$ (2 marks · state average 0.8 / 2, 40%)

- Uses $\int_0^{4/3} f(x)\,dx = 1$
- $m + \frac{4n}{3}$

Answer / notes: $m \cdot 1 + n \cdot \frac{4}{3} = m + \frac{4n}{3}$

TOPIC

The normal distribution

1 question, hardest first. None of this topic's questions made the hardest third, so this is the hardest one there is (★).

2023 VCAA Exam 2 · Section B · Question 4 ★

state average 49% of marks

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Tennis-ball diameters (normal), fitting through a container, grading, a confidence level, and serving-speed pdfs
Open the question in the official exam PDF (p. 18) ↗

a. Pr(D > 6.8) (4 d.p.) (1 mark · state average 0.8 / 1, 80%)

- ≈ 0.1587

Answer / notes: ≈ 0.1587
b. Diameter exceeded by only 10% of balls (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- ≈ 6.83 cm

Answer / notes: 90th percentile ≈ 6.83 cm
c. Pr(ball fits: D < 6.95) (4 d.p.) (1 mark · state average 0.8 / 1, 80%)

- ≈ 0.9938

Answer / notes: ≈ 0.9938
d. Pr(at least 3 of 4 fit) (4 d.p.) (2 marks · state average 1.3 / 2, 65%)

- Binomial with n = 4, p = 0.9938
- ≈ 0.9998

Answer / notes: ≈ 0.9998
e. Pr(grade A | fits) (4 d.p.) (2 marks · state average 1.2 / 2, 60%)

- $\frac{\Pr(6.54<D<6.86)}{\Pr(D<6.95)}$
- ≈ 0.8960

Answer / notes: ≈ 0.8960
f. σ so that more than 99% are grade A (2 d.p.) (2 marks · state average 0.7 / 2, 35%)

- $\frac{0.16}{\sigma} = 2.5758$
- $\sigma \approx 0.06$ cm

Answer / notes: Symmetric interval, so 0.5% in each tail.
g. Confidence level of the interval (0.7382, 0.9493) with n = 32 (2 marks · state average 0.5 / 2, 25%)

- $\hat{p} = 0.84375$, margin 0.10555, so $z \approx 1.645$
- 90%

Answer / notes: 90%. Many found z correctly but then said 95%.
h. Pr(grade A serving speed > 50) (4 d.p.) (1 mark · state average 0.6 / 1, 60%)

- ≈ 0.1345

Answer / notes: $\int_{50}^{3\pi^2+30} f(v)\,dv \approx 0.1345$
i. Exact mean serving speed of grade A balls (1 mark · state average 0.6 / 1, 60%)

- $3\pi^2 + 12$

Answer / notes: $E(V) = 3\pi^2 + 12$
j. a and b in $g(w) = af\left(\frac{w}{b}\right)$ if the mean is $2\pi^2 + 8$ (2 marks · state average 0.2 / 2, 10%)

- Area stays 1, so $ab = 1$; the mean scales by b: $b(3\pi^2 + 12) = 2\pi^2 + 8$
- $b = \frac{2}{3},\ a = \frac{3}{2}$

Answer / notes: $b = \frac{2}{3}$, $a = \frac{3}{2}$

TOPIC

Confidence intervals for proportions

1 question, hardest first.

2023 VCAA Exam 1 · Question 6

state average 35% of marks

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Working backwards from an approximate 95% confidence interval (0.04, 0.16)
Open the question in the official exam PDF (p. 7) ↗

a. $\hat{p}$ used (1 mark · state average 0.5 / 1, 50%)

- 0.1

Answer / notes: The midpoint: $\hat{p} = 0.10$.
b. Sample size (z = 2) (2 marks · state average 0.7 / 2, 35%)

- Margin $0.06 = 2\sqrt{\frac{0.1 \times 0.9}{n}}$
- $n = 100$

Answer / notes: $\sqrt{\frac{0.09}{n}} = 0.03 \Rightarrow n = 100$
c. Effect on the CI width of a sample four times as big (1 mark · state average 0.2 / 1, 20%)

- Width is halved (factor $\frac{1}{2}$)

Answer / notes: Width $\propto \frac{1}{\sqrt{n}}$, so $\times \frac{1}{2}$.

About this reference

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